1)

$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$

$CH_3Cl + Cl_2 \longrightarrow CH_2Cl_2 + HCl$

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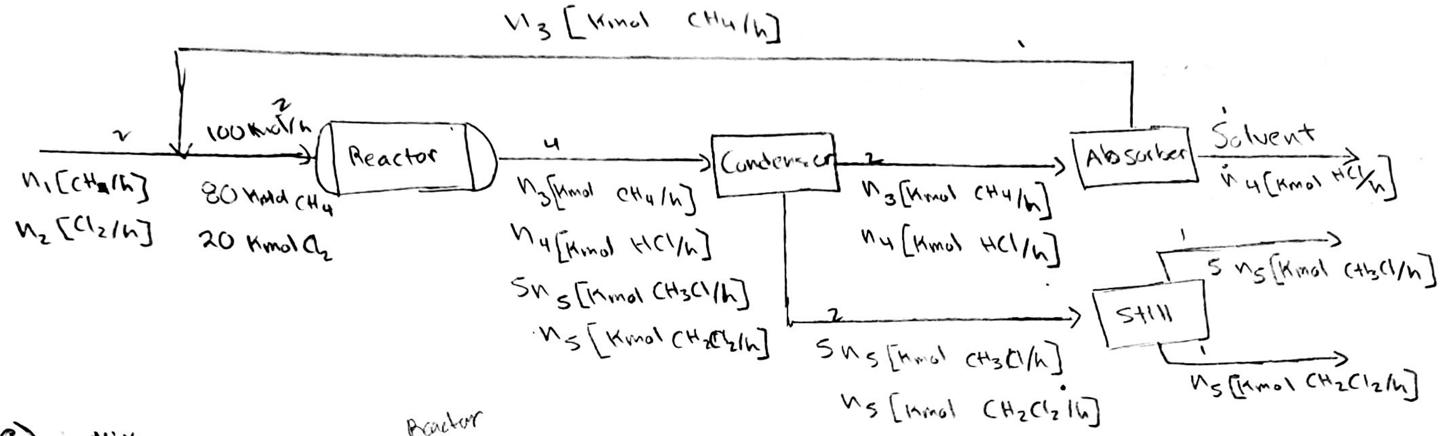
10-22-19
Homework 8
CHE 210



a) Mix

4a) 5　5a) 1
4b) 0　5b) 0
5c) 0
5d) 2

5 - 3 = 2

1

Reactor

4a) 8　5a) 1
4b) 2　5b) 1
5c) 0
5d) 5

8 - 7 = 1

0

$\frac{28}{40}$

Condenser

4a) 8　5a) 0
4b) 0　5b) 0
5c) 0
5d) 4

8 - 4 = 4

3

Absorber

4a) 4　5a) 0
4b) 0　5b) 0
5c) 0
5d) 2

4 - 2 = 0

2

Still

4a) 4　5a) 0
4b) 0　5b) 0
5c) 0
5d) 2

4 - 2 = 2

Overall

4a) 5　5a) 1
4b) 2　5b) 1
5c) 0
5d) 5

7 - 7 = 0

1

-5

1) 27
2) 30
3) 30
87

linearly dependent

$CH_4 = CH_{4\,out} = 80 - \xi_1$

$Cl_2 = 0 = 20 - \xi_1 - \xi_2$

$CH_3Cl = 5n_5 = \xi_1 - \xi_2$

$CH_2Cl_2 = n_5 = +\xi_2$

$HCl = n_4 = +\xi_1 + \xi_2$

$n_4 = 20$ Kmols/s

$5\xi_2 = \xi_1 - \xi_2$

$6\xi_2 = \xi_1$

$\xi_1 = 20 - \frac{1}{6}\xi_1$

$\xi_1 = 17.142$ Kmols

$\xi_2 = 2.8571$ Kmols

$CH_{4\,out} = n_3 = 62.858$ Kmols

$n_1 =$
n_2
n_3
n_4 = ?
n_5

c) 1000 kg/h of CH_3Cl = 19.8216 Kmols of CH_3Cl ✓

then 3.30 Kmols of CH_2Cl_2. So 23.1246 Kmols of Cl_2

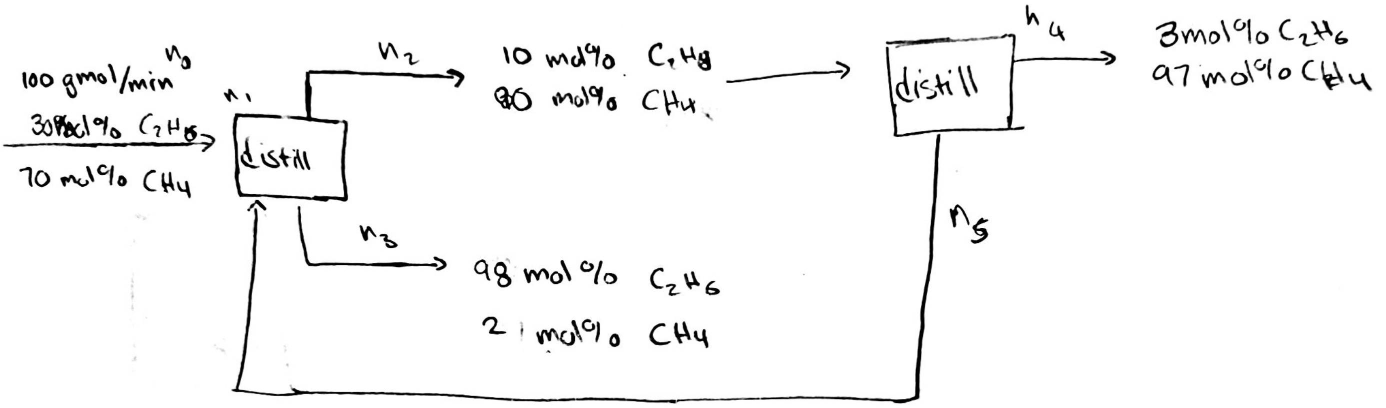
were put in, 92.4984 Kmols CH_4 and a total of 115.623 Kmol/h

The recycle stream being 69.3738 Kmols/h

Scale Factor = 1.87

-8

2) 5.32



dOF

4a) 10	5a) 2
4b) 0	5b) 4
	5c) 0
	5d) 4

10 - 10 = 0

For the Ind. Species Balance

$CH_{4_{out}} = CH_4 in - x_{3,CH_4}(n_3) - x_{5,CH_4}(n_5)$

$x_{4,CH_4}(n_4) = 70\ gmol/min - (.02)(n_3) - x_{5,CH_4}(n_5)$
$(.97)(n_4)$

$x_{3,C_2H_6}n_3 = C_2H_{6_{out}} = C_2H_6 in - x_{4,C_2H_6}n_4 - x_{5,C_2H_6}n_5$

$(.98)\,n_3 = \cancel{30\ gmol/min} - (.03)n_4 - x_{5,C_2H_6}n_5$

$CH_4 in\ \cancel{70} = .02n_3 + .97n_4 + x_{5,CH_4}(n_5)$

$C_2H_6 in\ \cancel{30} = .98n_3 + .03n_4 + (1 - x_{5,CH_4})(n_5)$

distillition on 2

CH_4: $x_{2,CH_4}(n_2) = x_{4,CH_4}(n_4) + x_{5,CH_4}(n_5)$

C_2H_6: $x_{2,C_2H_6}(n_2) = x_{4,C_2H_6}(n_4) + x_{5,C_2H_6}(n_5)$

$$\cancel{30\text{ mols} = .10(n_2) + .98(n_3)}$$
$$\cancel{70\text{ mol} = .90(n_2) + .02(n_3)}$$
$$\cancel{n_1 = n_2 + n_3}$$
$$\cancel{n_2 = \quad n_2 = 77.77\text{ mols} - .02\bar{2}\, n_3}$$
$$\cancel{30\text{ mols} = .1(77.77 - .022n_3) + .98n_3}$$
$$\cancel{30\text{ mols} = 7.77 - .0022n_3 + .98n_3}$$

2) $.9n_2 = .97n_4 + (1 - x_{5,C_2H_6})n_5$

$.1n_2 = .03n_4 + x_{5,C_2H_6}n_5$

$n_2 = n_4 + n_5$

$n_0 = n_2 + n_3$ $\qquad$ $n_1 = 200$ gmol/min

CH_4: $CH_{in} = x_{0,CH_4}(n_0) = 70 + x_{5,CH_4}n_5$

C_2H_6 in $= x_{0,C_2H_6}(n_0) = 30 + (1 - x_{5,CH_4})n_5$

$CH_{4\,out} = 70 - x_{3,CH_3}(n_3) - \cancel{x_{5,CH_4}(n_5)}$

$C_2H_{6\,out} = 30 - .02(n_3) - \cancel{x_{5,CH_4}(n_5)}$

$.97(n_4) = 70 - .02(n_3)$

$.98(n_3) = 30 - .03(n_4)$

$n_3 = 30.6122 - .03061(n_4)$

$.97(n_4) = 70 - .02(30.6122 - .03061\, n_4)$

$70 - .6000 - 6.122\times10^{-4}\, n_4$

$n_4 = 71.591$ gmols

$n_3 = 28.42$ gmols $\qquad$ $n_2 = 171.58$ gmols $\qquad$ $x_{2,CH_4} = 154.422$

$x_{2,C_2H_6} = 17.158$

$x_{4,CH_4} = 69.443$ gmols

$x_{4,C_2H_6} = 2.1477\,3$ $\qquad$ $x_{5,CH_4} = 84.979$ gmol

$x_{5,C_2H_6} = 15.021$ gmols

$n_4 = 71.592$ gmols

$n_3 = 28.42$ gmols

$x_{4,CH_4} = 69.442$ gmols

$x_{4,C_2H_6} = 2.14773$

$n_2 = 171.58$ gmols

$x_{5,CH_4} = 84.979$ gmols

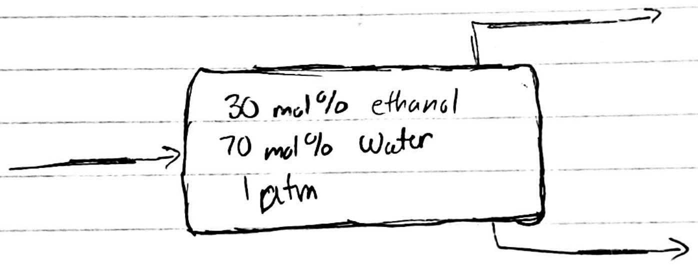
$x_{5,C_2H_6} = 15.021$ gmols

$\frac{71.58(.97)}{70} = .9919$ for fractional Yeild of methane

$\frac{30}{30}$

$\frac{2.842(.98)}{30} = .9284$ for fractional Yeild of Ethane

3)



Basis: 1 mol so .3 mol Eth

.7 mol H_2O

Flask Balence $F = V + L$

Ethane: $30 = 50y_{et} + 50x_{et}$

H_2O: $70 = 50y_{H_2O} + 50y_{H_2O}$

From the table You get

$\boxed{85.3^\circ C} \rightarrow 30 = 50(.4704) + 50(.1238)$ so close enough

3) Vapor: $y_{et} = .4704$ $y_{H_2O} = 1 - .4704 = .5296$

liquid: $x_{et} = .1238$ $x_{H_2O} = 1 - .1238 = .8762$

Vapor: $f_{R,et} = \frac{(.4704)50}{100(.30)} = .784$

liquid: $f_{R,H_2O} = \frac{(.8762)50}{100(.70)} = .6259$

Seperation factor $= \frac{x_{et,V}}{x_{H_2O,V}} \cdot \frac{x_{H_2O,L}}{x_{Et,L}} = 6.2864$

30/30